

Memory Architecture

FROM ARRAYS TO LINKED LISTS

DSA Module: Memory Management

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1. Principles of Storage

Computer memory (RAM) is a series of addressable bytes. Every variable you create occupies a specific street address in this memory space.

The Array Constraint: Contiguous Memory

Arrays must be stored in **contiguous blocks**. This means elements are placed exactly one after another. If you have a 4-element integer array, they are locked into a single row.

Val 3

Val 5

Val 13

Val 2

Problem: If you remove "Val 5", you must shift Val 13 and Val 2 to close the gap. This is slow.

2. Dynamic Distribution

Linked Lists break the contiguous requirement. Nodes can be scattered anywhere in memory where there is a free gap.

The Node structure (\$C\$ Perspective)

On an 8-bit microcontroller, a node might look like this:

- **Bytes 1-2:** Data (The Value)

- **Bytes 3-4:** Address (Where is the next node?)



Notice the "Gaps". Because each node points to the next, we don't care about shifting other elements when we add a new one.

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3. Implementation Deep Dive

C-Language (Manual Memory)

```
typedef struct Node {
    int data;
    struct Node* next;
} Node;

Node* createNode(int data) {
    Node* newNode = (Node*)malloc(sizeof(Node));
    newNode->data = data;
    newNode->next = NULL;
    return newNode;
}
```

Python (Abstract Memory)

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

node1 = Node(3)
node2 = Node(5)
node1.next = node2 # Internal address linking
```

4. The Memory Verdict

Benefits:

- **No Shifting:** Inserting/removing elements doesn't require moving other data.
- **Gap Utilization:** Can use fragmented memory that an array can't touch.

Downsides:

- **Memory Overhead:** Every node needs extra space just to store the address of the next node.
- **No Direct Access:** You can't jump to the 5th element; you must follow the pointer breadcrumbs from the start.

5. Knowledge Check

Exercise: When you remove a node from a Linked List, do you have to shift the other nodes in memory?

Answer: NO

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